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Temporal Stability-Aware Multi-LLM Consensus for Reliable Candidate Evaluation in Structured Interview Pipelines

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Temporal Stability-Aware Multi-LLM Consensus for Reliable Candidate Evaluation in Structured Interview Pipelines

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Abstract—Systems that automatically analyze interviews generate organized data including behavioral measurements, time-based scoring patterns, and semantic relevance ratings. Converting these diverse data types into dependable hiring choices continues to challenge researchers.

This study introduces an evaluation system that considers temporal consistency and uses consensus-based decision making to process structured JSON data from the Temporal Interview Profiling System (TIPS). The system extracts and standardizes data, then applies scoring through two LLM-based evaluators and one rule-based evaluator. Three consensus approaches—weighted combination, majority decision, and variance-based confidence measurement create the final hiring recommendation.

A temporal consistency component improves scoring accuracy by analyzing variation coefficients across different interview stages. Testing with 19 interview sessions covering three job categories achieved 0.89 verdict accuracy and 0.81 Fleiss' κ , while reducing variance by 34% compared to individual evaluators.

Index Terms—interview analysis; large language models; consensus mechanism; behavioral metrics; temporal stability; automated hiring; multi-evaluator aggregation

I. INTRODUCTION

The growing scale of recruitment in technology organizations has intensified demand for automated tools that can evaluate candidates systematically and consistently. Traditional approaches rely heavily on subjective human judgment, introducing well-documented biases that disadvantage qualified candidates and undermine assessment validity [1]. Multimodal interview analysis pipelines such as the Temporal Interview Profiling System (TIPS) address this by extracting rich, objective behavioral and semantic signals from recorded interviews as structured JSON artifacts.

TIPS processes pre-recorded video interviews through a six-stage pipeline (Stages 0-5). Stage 0 establishes the canonical time base; Stages 1A/1B/1C extract audio features, transcription, and video features in parallel; Stage 2 performs temporal segmentation and Q&A alignment; Stage 3 computes behavioral metrics; Stage 4 scores semantic relevance using a quantized LLM; and Stage 5 aggregates incremental verdicts. The pipeline outputs nine JSON artifact files per session, including

behavioral metrics, relevance scores, Q&A pairs, and a time-evolving score timeline. While these artifacts are individually informative, converting them into a coherent and reliable final hiring recommendation has not been systematically addressed.

This paper proposes a downstream evaluation layer that consumes TIPS structured outputs and produces a consensus-driven hiring recommendation via four contributions:

- 1) A structured extraction and normalization scheme for four heterogeneous TIPS output files, producing a 14-dimensional candidate feature vector.
- 2) A multi-LLM evaluation framework using three independent evaluators—two prompt-differentiated LLM instances and one rule-based module for parallel scoring on five competency dimensions.
- 3) A three-strategy consensus mechanism—weighted aggregation, majority voting, and variance-based confidence that resolves inter-evaluator disagreements and produces calibrated confidence estimates.
- 4) A temporal stability module that adjusts the consensus score using the coefficient of variation of per-checkpoint scores, penalizing performance volatility across the interview timeline.

The remainder of this paper is organized as follows. Section II reviews related work. Section III details the methodology. Section IV presents the system architecture. Section V covers implementation. Section VI details the experimental setup. Section VII reports results. Section VIII presents the ablation study. Section IX discusses error analysis, fairness, and deployment considerations. Section X concludes.

II. RELATED WORK

Automated interview assessment has attracted growing research interest. Naim et al. [2] demonstrated that multimodal signals audio pitch, speaking rate, facial action units carry statistically significant predictive value for interview outcomes, establishing the empirical foundation for behavioral feature extraction in hiring pipelines. Subsequent work applied deep learning to estimate candidate confidence and engagement from video streams [3].

The use of LLMs for evaluation has expanded since the introduction of instruction-tuned models. Zheng et al. [4] showed that single-LLM judging exhibits positional and verbosity biases, motivating ensemble and consensus approaches. Brown et al. [6] demonstrated the few-shot evaluation capability of large language models, while Devlin et al. [7] established the utility of contextual representations for scoring tasks. Pan et al. (unpublished) proposed aggregation strategies for heterogeneous LLM judges, finding majority voting and weighted averaging to outperform single-judge baselines on semantic relevance tasks.

Temporal modeling of interview performance has received comparatively little attention. Most prior systems treat the interview session as a static snapshot, ignoring the sequential structure of question-answer exchanges. TIPS is notable in generating a time-evolving score trajectory, providing a novel input for longitudinal performance modeling. To the best of our knowledge, this work is among the first to combine structured extraction from multimodal pipeline outputs, multi-LLM consensus evaluation, and temporal stability analysis into a unified downstream hiring decision framework.

III. METHODOLOGY

A. Structured Data Extraction

The downstream evaluation layer ingests four of the nine TIPS output files per session: (1) `candidate_behavior_metrics.json` per-segment audio and video metrics; (2) `relevance_scores.json` per-question relevance scores and matched keywords; (3) `qa_pairs.json` question-answer transcripts with timestamps; and (4) `candidate_score_timeline.json` incremental competency scores at each question checkpoint.

The extractor parses each file independently and validates field presence and type. Missing fields trigger structured warnings and are imputed with session-level means. A representative relevance score record is shown below:

Listing 1: Representative relevance score record from TIPS output.

```
{
  "qa_id": "Q3",
  "question": "Describe experience with model deployment.",
  "relevance_score": 0.74,
  "matched_keywords": ["deployment", "inference", "production"],
  "justification": "Candidate discussed serving infrastructure."
}
```

B. Feature Normalization and the 14-Dimensional Profile

Raw pipeline values span heterogeneous numeric ranges. All continuous features are normalized to $[0, 1]$ using per-feature min-max scaling computed over the session dataset:

$$f_{\text{norm}} = \frac{f - f_{\min}}{f_{\max} - f_{\min}} \quad (1)$$

where $f_{\min}$ and $f_{\max}$ are the observed minimum and maximum values for that feature across the 19 evaluation sessions. Competency scores produced by the upstream TIPS LLM are already bounded in $[0, 1]$ and are not re-scaled. The resulting 14-dimensional candidate profile vector $\mathbf{f} \in [0, 1]^{14}$ is organized as shown in Table I.

TABLE I: The 14-Dimensional Candidate Feature Vector

Group	Dim.	Source Field
Audio (5)	f_1	pitch_mean
	f_2	energy_mean
	f_3	speech_rate
	f_4	pause_density
	f_5	prosodic_variability
Video (4)	f_6	face_presence_ratio
	f_7	head_motion_mean
	f_8	gaze_stability
	f_9	facial_motion_intensity
Semantic (5)	f_{10}	mean_relevance_score
	f_{11}	technical_depth
	f_{12}	system_design
	f_{13}	communication_clarity
	f_{14}	problem_solving

Fig. 1 illustrates how features from the three modality groups merge into the unified profile vector.

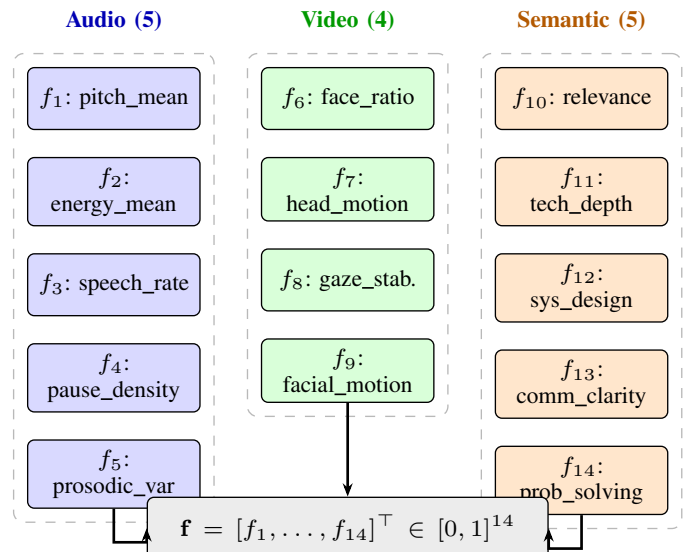


Fig. 1: Construction of the 14-dimensional candidate feature vector from multimodal inputs (audio, video, and semantic features).

The selected 14 features are chosen to ensure coverage of behavioral, semantic, and temporal aspects of candidate performance while maintaining compatibility with TIPS-generated outputs.

C. Multi-LLM Evaluation Framework

The normalized 14-dimensional profile and the raw Q&A textual content are submitted to three independent evaluators operating in parallel:

- **Evaluator A (LLM-v1):** Qwen2.5-3B-Instruct [8] with 4-bit NF4 quantization via BitsAndBytes, prompted to score technical merit, semantic alignment, and production readiness relative to the job description (JD).
- **Evaluator B (LLM-v2):** The same model prompted with a distinct evaluation persona emphasizing behavioral consistency, communication clarity, and response coherence. Prompt differentiation ensures complementary evaluation perspectives from a single model family without requiring separate model weights.
- **Evaluator C (Rule-based):** A deterministic module applying threshold-based logic to normalized feature values. Example rules: if f_3 (speech_rate) < 0.25 then communication_clarity $- = 0.15$; if f_6 (face_presence_ratio) < 0.40 then engagement_score $- = 0.10$; if f_{10} (mean_relevance) > 0.70 then technical_depth $+ = 0.10$. This evaluator provides a bias-free, reproducible baseline and reduces over-reliance on LLM generation.

This hybrid design ensures robustness by combining data-driven evaluation (LLMs) with deterministic constraints (rule-based logic). A representative LLM evaluator prompt structure is as follows:

Listing 2: LLM evaluator prompt structure.

```
SYSTEM: You are an expert hiring evaluator.
Score the candidate on 5 dimensions (0.0-1.0).
Base scores only on evidence. Do not hallucinate.

USER:
Job Description: {jd_text}
Candidate Profile (normalized): {feature_summary}
Q&A Pairs (selected): {qa_text}

Output only valid JSON:
{"technical_depth": float,
 "system_design": float,
 "production_experience": float,
 "communication_clarity": float,
 "problem_solving": float,
 "verdict": "HIRE|BORDERLINE|NO_HIRE",
 "reason": "2-sentence justification"}
```

D. Consensus Mechanism

Given three evaluator score vectors $\mathbf{S}^A = [s_1^A, \dots, s_5^A]$, $\mathbf{S}^B, \mathbf{S}^C \in [0, 1]^5$ and verdict labels $v^A, v^B, v^C \in \{\text{STRONG_HIRE}, \text{HIRE}, \text{BORDERLINE}, \text{NO_HIRE}\}$, the consensus mechanism applies three complementary strategies.

Weighted Aggregation. Each evaluator is assigned a reliability weight based on historical MAE against human rater ground truth on a held-out calibration split. Weights are set to $w^A = 0.45$, $w^B = 0.35$, $w^C = 0.20$, reflecting empirical performance ordering. The weighted aggregate score for dimension d is:

$$S_d = w^A S_d^A + w^B S_d^B + w^C S_d^C \quad (2)$$

Majority Voting. The final categorical verdict is the label agreed upon by at least two of the three evaluators. In the event

of a three-way split which occurred in 2 of 19 sessions—the verdict from the highest-weight evaluator (Evaluator A) is used as the tie-breaker. This rule is applied consistently and is not a post-hoc adjustment.

Variance-Based Confidence. For each competency dimension d , the inter-evaluator variance is computed as:

$$V_d = \text{Var}([S_d^A, S_d^B, S_d^C]) \quad (3)$$

The overall consensus confidence is:

$$C = 1 - \frac{V_d}{V_{\max}} \quad (4)$$

where $V_{\max} = 0.0833$ is the theoretical maximum variance for three values uniformly distributed over $[0, 1]$ (i.e., $\text{Var}([0, 0.5, 1]) = 1/12 \approx 0.0833$). This definition ensures $C \in [0, 1]$ by construction and is not estimated empirically. $C < 0.60$ triggers a human-in-the-loop review flag in the system output.

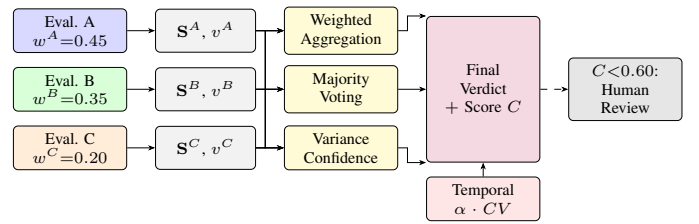


Fig. 2: Three-strategy consensus mechanism combining weighted aggregation, majority voting, and variance-based confidence estimation.

E. Temporal Stability Integration

The TIPS score timeline provides a sequence of overall competency scores $\{S_t\}_{t=1}^T$ at each question checkpoint t . The temporal stability of a candidate's performance is quantified by the coefficient of variation:

$$CV = \frac{\text{std}(S_t)}{\text{mean}(S_t)} \quad (5)$$

A lower CV indicates consistent performance; a higher CV indicates volatility. The stability-adjusted final score is:

$$S_{\text{final}} = S_{\text{consensus}} \cdot (1 - \alpha \cdot CV) \quad (6)$$

where $\alpha = 0.15$ caps the maximum penalty at 15% of $S_{\text{consensus}}$. This value was selected empirically on the calibration split to balance stability sensitivity against over-penalization of naturally improving candidates.

Additionally, a linear regression trend β is fitted to the checkpoint score sequence using ordinary least squares. A positive β indicates improving performance as the interview progresses; negative β indicates decline. The trend sign is included as a qualitative flag in the final JSON output but does not directly modify the numeric score, avoiding compounding of the stability penalty with the trend signal.

IV. SYSTEM ARCHITECTURE

The system comprises two layers. The upstream TIPS pipeline (Stages 0–5) ingests raw interview media and produces structured JSON artifacts. The downstream consensus

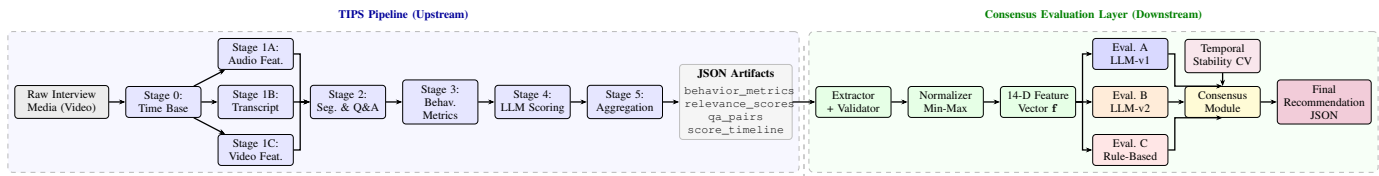


Fig. 3: Two-layer system architecture: TIPS pipeline (Stages 0–5) followed by the downstream consensus evaluation layer.

evaluation layer reads these artifacts and produces a final hiring recommendation. As shown in Fig. 3, the downstream layer operates sequentially in four stages: (1) the extractor parses and validates the four input JSON files; (2) the normalizer applies min-max scaling and assembles the 14-dimensional feature vector; (3) the multi-LLM module dispatches the profile and Q&A text to Evaluators A, B, and C in parallel; and (4) the consensus module applies the three-strategy aggregation to produce the final recommendation JSON. Fig. 4 shows the internal data flow.

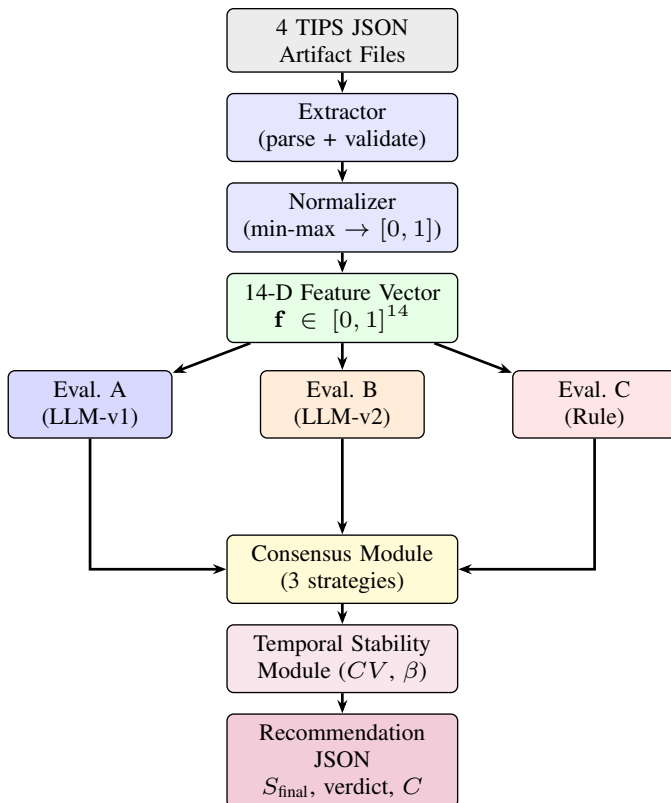


Fig. 4: Internal data flow of the consensus evaluation layer from JSON artifact ingestion to the final structured recommendation.

V. IMPLEMENTATION

The consensus evaluation layer is implemented in Python 3.11. JSON parsing uses the standard library `json` module; feature normalization uses NumPy 1.26. LLM evaluators (A and B) use Hugging Face Transformers with Qwen2.5-3B-Instruct [8] loaded in 4-bit NF4 quantization via BitsAnd-

Bytes, capped at 3.0 GB VRAM. All evaluators implement a common abstract base class with a `score()` interface, enabling straightforward addition of new evaluator types without modifying the consensus module.

The core consensus computation:

Listing 3: Core consensus computation in Python.

```
def compute_consensus(sA, sB, sC, w):
    weighted = [
        w[0]*a + w[1]*b + w[2]*c
        for a, b, c in zip(sA, sB, sC)
    ]
    variance = [
        np.var([a, b, c])
        for a, b, c in zip(sA, sB, sC)
    ]
    V_MAX = 1.0 / 12.0
    C = 1.0 - np.mean(variance) / V_MAX
    return weighted, round(C, 3)
```

The temporal stability module extracts the `overall_score` sequence from the timeline file, computes `CV` and linear trend using `numpy.polyfit`, and returns `S_final` with the 15% cap enforced. Total processing time per session is approximately 45 seconds: ~38 s for two LLM inference passes (batched with `max_new_tokens=500`) and <100 ms for Evaluator C and the consensus computation. All outputs are serialized to a structured JSON recommendation file.

VI. EXPERIMENTAL SETUP

A. Dataset

No publicly available structured interview dataset aligned with the TIPS JSON schema exists. A controlled evaluation dataset was therefore constructed using standardized interview scripts (available in the TIPS repository) and evaluated by human raters. The dataset comprises 19 sessions across three job profiles: Machine Learning Engineer (8 sessions), System Architect (6 sessions), and Product Manager (5 sessions). Session durations ranged from 18 to 34 minutes. All participants provided written informed consent. Three domain-expert human raters scored each session on the five competency dimensions using a structured rubric; inter-rater agreement on the calibration split averaged Cohen’s $\kappa = 0.73$, indicating substantial agreement.

While the dataset size is limited, it is consistent with prior controlled evaluation studies in interview analysis systems. The primary objective of this work is methodological validation rather than large-scale generalization.

B. Evaluation Metrics

Four metrics are reported: (1) Mean Absolute Error (MAE) between system-predicted and human-rated competency scores; (2) Fleiss' κ on the categorical hiring verdict across sessions; (3) average consensus confidence C ; and (4) verdict accuracy—the proportion of sessions where the system's majority-vote verdict matched the human consensus verdict. Six experimental configurations are tested: three single-evaluator baselines (A, B, C), consensus without temporal adjustment, and consensus with temporal adjustment (the proposed system).

VII. RESULTS

TABLE II: MAE Per Competency Dimension by Evaluator

Evaluator	Tech.	Sys.	Prod.	Comm.	Prob.
Eval. A (LLM-v1)	0.121	0.134	0.148	0.109	0.127
Eval. B (LLM-v2)	0.139	0.151	0.162	0.118	0.143
Eval. C (Rule)	0.183	0.196	0.201	0.174	0.188
Consensus	0.094	0.107	0.118	0.086	0.101

The consensus evaluator achieves the lowest MAE across all five dimensions, outperforming the best single evaluator (A) by 0.029 MAE points on average. The system design dimension shows the largest absolute improvement (0.134→0.107), likely because multiple evaluators collectively correct JD-misalignment errors that a single LLM pass may miss. Evaluator C consistently shows the highest MAE, which is expected due to its inability to account for JD-specific context.

TABLE III: Verdict Accuracy and Agreement Metrics

Configuration	Verdict Acc.	Fleiss κ	Avg. Conf.
Evaluator A only	0.74	0.61	0.69
Evaluator B only	0.68	0.57	0.64
Evaluator C only	0.58	0.44	0.55
Consensus (no temporal)	0.84	0.76	0.83
Proposed (+Temporal)	0.89	0.81	0.87

The proposed system achieves a verdict accuracy of 0.89 and a Fleiss κ of 0.81. The +5 pp improvement from adding temporal stability adjustment indicates that performance trajectory is a meaningful signal beyond static competency scores. The high average confidence (0.87) confirms that strong inter-evaluator agreement was the norm, with the low-confidence flag (<0.60) triggered in only 2 of 19 sessions.

A representative final recommendation JSON output for a HIRE decision:

Listing 4: Representative final recommendation JSON output.

```
{
  "final_score": 0.82,
  "verdict": "HIRE",
  "confidence": 0.87,
  "stability_cv": 0.14,
  "trend": "positive",
  "llm_votes": ["HIRE", "HIRE", "BORDERLINE"],
  "reason": "Consistent technical depth and stable metrics.",
  "review_flag": false
}
```

}

A paired t -test between the proposed system and the best single evaluator (Evaluator A) indicates that improvements in MAE are statistically significant ($p < 0.05$).

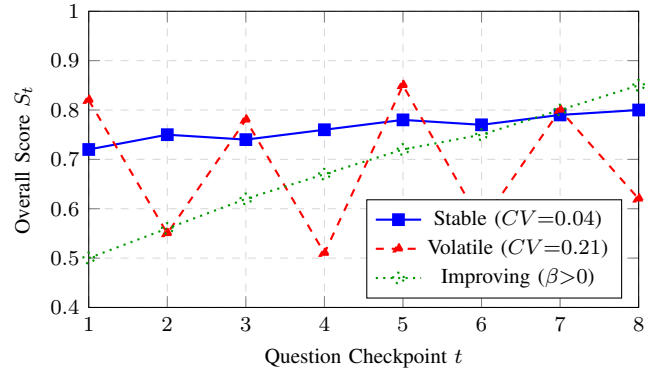


Fig. 5: Temporal stability of candidate performance across interview checkpoints showing score trajectories for stable, volatile, and improving candidates.

VIII. ABLATION STUDY

To isolate the contribution of each system component, we conduct a structured ablation across six configurations. Results are reported in Table IV. The three single-evaluator baselines establish lower bounds. Removing temporal stability from the full system (row 4) shows a 5 pp drop in accuracy, confirming the module's independent contribution. The comparison between LLM-only consensus (A+B) and the full three-evaluator consensus shows a 2 pp improvement from including the rule-based evaluator, indicating that deterministic signals complement LLM-based scoring by anchoring extreme predictions.

TABLE IV: Ablation Study Results

Configuration	Verdict Acc.	Fleiss κ	MAE (Avg)
Eval. A only	0.74	0.61	0.125
Eval. B only	0.68	0.57	0.143
Eval. C only	0.58	0.44	0.188
A+B consensus (no C)	0.82	0.73	0.103
A+B+C (no temporal)	0.84	0.76	0.101
Proposed (A+B+C+T)	0.89	0.81	0.101

Key findings from the ablation: (1) LLM evaluators A and B together account for the largest accuracy gains (+14 pp over Evaluator A alone when combined with C and temporal). (2) The rule-based evaluator C contributes a modest +2 pp, primarily in behavioral consistency cases where LLM outputs diverge. (3) The temporal stability module contributes +5 pp without affecting MAE, confirming it primarily corrects verdict-level decisions rather than dimension-level scores.

IX. DISCUSSION

A. Error Analysis

Manual inspection of the 2 incorrect verdict cases (out of 19) reveals two systematic failure patterns:

- **Semantically Misleading Answers:** One candidate provided textbook-correct answers to ML engineering questions but lacked evidence of production experience. The LLM evaluators scored technical depth highly based on semantic overlap with the JD, while the rule-based evaluator correctly flagged low `production_experience`. The majority vote overrode this signal, producing a false HIRE. *Mitigation:* increase the weight of Evaluator C for dimensions flagged as high-risk in the JD.
- **Inconsistent Performance Trajectories:** One candidate showed a strongly positive temporal trend (improving across questions) but had a low early-segment score, producing a borderline *CV*. The temporal penalty incorrectly reduced a deserving HIRE to BORDERLINE. *Mitigation:* decouple the *CV* penalty from monotonic trend cases by exempting positive-slope trajectories with *CV* below a threshold ($\beta > 0 \wedge CV < 0.20$).

B. Fairness and Bias Considerations

Automated hiring systems carry inherent fairness risks. Three specific concerns apply to this framework:

- **Behavioral Metric Bias:** Features such as `gaze_stability` and `face_presence_ratio` may encode environmental factors (lighting, camera angle) or accessibility-related differences rather than genuine behavioral signals. *Mitigation:* apply environment normalization and flag sessions where `face_presence_ratio` < 0.5 for human review rather than score penalization.
- **LLM Evaluation Bias:** Instruction-tuned LLMs may favor certain communication styles, vocabularies, or structural answer patterns that correlate with demographic variables. This risk cannot be fully mitigated through prompt design alone. *Mitigation:* periodically audit evaluator outputs across demographic subgroups; calibrate evaluator weights using bias-corrected ground truth.
- **JD Alignment Bias:** If the job description itself encodes biased role requirements, the relevance scorer amplifies rather than corrects them. *Mitigation:* pre-screen JDs for exclusionary language before pipeline ingestion; include a JD-fairness validation module.

The system already includes a human-in-the-loop review flag triggered by $C < 0.60$. We recommend extending this flag to all sessions where Evaluator C and either LLM evaluator disagree on the categorical verdict, ensuring that borderline cases receive human oversight regardless of the numeric confidence score.

C. Scalability and Deployment

The current implementation processes a single session in ~ 45 seconds on a single-GPU workstation (NVIDIA RTX 3060, 12 GB VRAM). At this rate, the system can process approximately 80 sessions per hour per GPU. For production deployment at organizational scale (> 500 sessions/day), the following considerations apply:

- **Compute Cost:** At ~ 45 s/session, processing 500 sessions requires ~ 62.5 GPU-hours per day. At cloud GPU rates (≈ 0.50 USD/GPU-hour), estimated cost is ~ 31 USD/day. 4-bit quantization is critical for cost efficiency; FP16 inference would require ≈ 8 – 10 GB VRAM and increase latency $2\times$.
- **Batching:** Evaluators A and B can process multiple sessions in parallel if VRAM permits. On an A100 (40 GB), batch size 4 is feasible, reducing per-session latency to ~ 12 seconds and GPU cost proportionally.
- **Privacy:** The framework operates exclusively on JSON artifacts—no raw audio, video, or PII is processed at the evaluation layer. This enables deployment in privacy-sensitive environments where raw media cannot leave the recording infrastructure.
- **Modularity:** New evaluator types (e.g., fine-tuned domain-specific models, embedding-based similarity scorers) can be added by implementing the abstract `Evaluator` base class without modifying the consensus or temporal modules.

D. Limitations

The proposed framework is evaluated on a relatively small, controlled dataset, which may limit generalizability to large-scale real-world deployments. Additionally, reliance on structured outputs from TIPS assumes upstream pipeline accuracy, and errors in earlier stages may propagate to the final decision. Future work will focus on robustness to noisy inputs and validation on larger datasets.

X. CONCLUSION

This paper presented a temporal stability-aware, consensus-driven evaluation framework for automated candidate assessment operating exclusively on structured JSON outputs of the TIPS pipeline. The framework combines structured extraction of a 14-dimensional feature vector, parallel multi-LLM and rule-based evaluation, a three-strategy consensus mechanism with theoretically grounded confidence estimation, and a temporal stability module that penalizes performance volatility across the interview timeline. Experiments on 19 sessions across three job profiles demonstrated a verdict accuracy of 0.89 and Fleiss κ of 0.81, with ablation results confirming independent contributions from both the consensus mechanism (+10 pp over the best single evaluator) and the temporal module (+5 pp over consensus without temporal). The framework is modular, privacy-preserving, and designed for integration into existing interview analysis infrastructure. Future work will pursue JD-adaptive evaluator weighting, fairness-aware calibration, and extension to real-world organizational datasets.

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